

Extracting the quantum geometric tensor of an optical Raman lattice by Bloch-state tomography

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In Hilbert space, the geometry of the quantum state is identified by the quantum geometric tensor (QGT), whose imaginary part is the Berry curvature and whose real part is the quantum metric tensor. Here, we experimentally realize a complete Bloch-state tomography to directly measure the eigenfunction of an optical Raman lattice for ultracold atoms. Through the measured eigenfunction, the distribution of the complete QGT in the Brillouin zone is reconstructed, with which the topological invariants are extracted by the Berry curvature and the distances of quantum states in momentum space are measured by the quantum metric tensor. Furthermore, we experimentally test a predicted inequality between the Berry curvature and the quantum metric tensor, which reveals a deep connection between the topology and geometry.

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In quantum physics, quantum states are defined in Hilbert spaces, in which Berry phases profoundly reflect the fundamental geometric structure of Hilbert spaces [1]. In 1989, Berry further pointed out that the geometry of the Hilbert space is fully determined by the quantum geometry tensor (QGT) [2]. The QGT has an antisymmetric imaginary part, i.e., the well-known Berry curvature, which unveils the geometric phase difference of a wave function between two parameter points in the parameter space. Its associated topological phenomena have been extensively studied in many systems, such as the Aharonov-Bohm effect [3], the quantum Hall effect [4], and topological insulators and topological superconductors [5,6]. Moreover, the QGT has a symmetric real part termed the quantum metric tensor [7,8], which describes the distance between two quantum states in the parameter space. The quantum metric tensor can give the famous Euler characteristic number [9] and is of great importance in obtaining the quantum Fisher information [10,11] as well as depicting quantum phase transitions [12].

The broadening of topological knowledge from condensed matter physics has rendered the possibility of achieving quantum control with a view towards measuring the QGT. Recently, several experiments have been conducted to measure

the QGT of two-level systems in a superconducting qubit [13], a nitrogen-vacancy (NV) center qubit [14,15], and exciton-polaritons [16]. Nevertheless, in lattice systems, the complete quantum geometric tensor of Bloch bands is yet to be measured. Specially, in ultracold atoms, optical lattices have been developed as a highly controllable and tunable quantum simulator to investigate topological band physics [17], which offers an ideal platform for the measurement of the QGT [18]. In addition, there are several tomography experiments in cold-atom systems, based on the dynamics after a projection onto flat bands [19–21], based on projection on different momenta by fast acceleration [22], based on the interference of paths surrounding [23] or crossing band closing points [24], and based on off-resonant coupling to higher bands [25]. Whereas these tomography protocols probe different aspects of Bloch states in optical lattices, none of them focus on studying the quantum geometric tensor.

Here, we experimentally measure the complete QGT of a topological band using ultracold atoms in an optical lattice. We implement a complete Bloch-state tomography using a well-controlled momentum-transferred Raman laser pulse in a two-dimensional (2D) optical Raman lattice [26,27]. Based on the Bloch-state tomography, the eigenfunction of the two-band model in the Raman lattices is directly measured through the expectation values of three Pauli matrices. Thus the momentum-resolved Berry curvature and the quantum metric tensor, relying only on the eigenfunction, are simultaneously reconstructed, which probes the complete geometry of Bloch states in the Raman lattices. Meanwhile, the Chern number [5] (quantum volume [28,29]) is obtained from the integral of the Berry curvature (the quantum metric tensor) over

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the full Brillouin zone. An inequality proved in Refs. [28,29] between the quantum volume and the Chern number is also tested experimentally.

Our Bloch-state tomography is realized in a 2D Raman lattice with ultracold ^{87}Rb atoms [27,30,31], as shown in Fig. 1(a). The Raman lattices are constructed by two laser beams $E_{x,y}$ with wavelength $\lambda = 787$ nm that propagate in opposite directions as well as couple two magnetic levels $|F = 1, m_F = -1\rangle \equiv |\uparrow\rangle$ and $|F = 1, m_F = 0\rangle \equiv |\downarrow\rangle$. The Hamiltonian is ($\hbar = 1$)

$$H = \frac{k^2}{2m} + V_{\text{latt}} + \frac{\delta}{2}\sigma_z + \Omega_1\sigma_x + \Omega_2\sigma_y, \quad (1)$$

where $\mathbf{k} = (k_x, k_y)$, m , δ , and $\sigma_{x,y,z}$ are the momentum, the mass of the atom, the two-photon detuning, and the Pauli matrices, respectively. The Raman potential $\Omega_{1,2}$ (lattice potential V_{latt}) induces spin-flipped (spin-conserved) hopping with strength Ω_0 (V_0) and transfers momentum of $\mathbf{k}_0 = (\pm k_0, \pm k_0)$ ($2\mathbf{k}_0$), which is observed from the atomic distribution of the ground state in the momentum space [Fig. 1(b)] [26,27]. Then, the eigenstate $|\Psi(\mathbf{q})\rangle$ of Hamiltonian equation (1) can be denoted as the superposition of $|\mathbf{q}, \uparrow\rangle$ and $|\mathbf{q} + \mathbf{k}_0, \downarrow\rangle$, i.e., $|\Psi(\mathbf{q})\rangle = c_\uparrow(\mathbf{q})|\mathbf{q}, \uparrow\rangle + c_\downarrow(\mathbf{q})|\mathbf{q} + \mathbf{k}_0, \downarrow\rangle$ [31]. Here, $\mathbf{q} = (q_x, q_y)$ and $c_{\uparrow,\downarrow}$ are the quasimomentum and the normalized complex coefficient, respectively. The eigenstate $|\Psi(\mathbf{q})\rangle$ can be represented on the Bloch sphere, whose poles are $|\mathbf{q}, \uparrow\rangle$ and $|\mathbf{q} + \mathbf{k}_0, \downarrow\rangle$ [Fig. 1(c)].

The main idea of the tomography is to measure the expectation values of three Pauli matrices $\langle\sigma_{x,y,z}(\mathbf{q})\rangle$ for Raman lattices by rotating the measurement basis (see Supplemental Material [31] for details). We directly obtain the expectation value $\langle\sigma_z(\mathbf{q})\rangle$ by spin-resolved time-of-flight (ToF) imaging, which has been applied in Refs. [26,27]. After rotating the measurement basis $|\mathbf{q}, \uparrow\rangle$ and $|\mathbf{q} + \mathbf{k}_0, \downarrow\rangle$ to the x axis (y axis) on the Bloch sphere, the expectation value $\langle\sigma_x(\mathbf{q})\rangle$ ($\langle\sigma_y(\mathbf{q})\rangle$) is observed by spin-resolved ToF imaging [32]. Such rotation is achieved by a coherent Raman pulse. The pulse acting on $|\Psi(\mathbf{q})\rangle$ requires transferring the momentum $\mathbf{k}_0 = (-k_0, -k_0)$ between $|\mathbf{q}, \uparrow\rangle$ and $|\mathbf{q} + \mathbf{k}_0, \downarrow\rangle$ as well as maintaining a fixed relative phase $\Delta\varphi$ between the Raman pulse and the Raman lattices, which ensures accurate determination of $\langle\sigma_{x,y}(\mathbf{q})\rangle$ [Fig. 1(c)].

Concretely, the Raman pulse is made up of incident laser beams of $E_{x,y}$ with retroreflective beams switched off by acousto-optic modulators (AOMs) [Fig. 1(a)]. Thus the Hamiltonian of the Raman pulse is [31]

$$H_R = \begin{pmatrix} k^2/2m + \delta_R/2 & \Omega_R \\ \Omega_R^* & k^2/2m - \delta_R/2 \end{pmatrix}. \quad (2)$$

In the experiment, a sufficient short Raman pulse with two-photon detuning $\delta_R = 0$ is applied such that the kinetic energy is ignored and Raman coupling Ω_R dominates. $\Omega_R = \Omega_{R0}e^{-i[k_0(x+y) - \Delta\varphi]}$ with strength Ω_{R0} launches a momentum transfer of $\mathbf{k}_0 = (-k_0, -k_0)$ between $|\mathbf{q}, \uparrow\rangle$ and $|\mathbf{q} + \mathbf{k}_0, \downarrow\rangle$. The relative phase $\Delta\varphi$ is controlled by the initial phase difference of the input lasers $E_{x,y}$ that generates the Raman pulse and the Raman lattices [31].

When the Raman pulse is applied to $|\Psi(\mathbf{q})\rangle$ with the duration time t_R , the relative phase $\Delta\varphi$ is imprinted onto the time-dependent state $|\Psi(\mathbf{q}, t_R)\rangle$. The state $|\Psi\rangle$ precesses

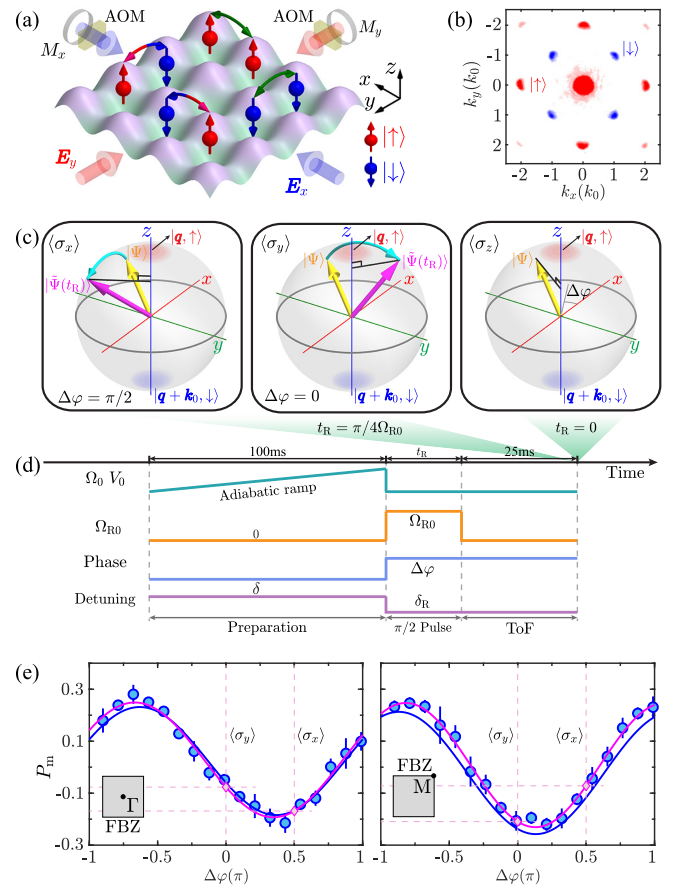


FIG. 1. Bloch-state tomography in 2D optical Raman lattices. (a) Sketch map of Raman lattices. (b) The distribution of Bose-Einstein condensates (BECs) in momentum space. The BEC is prepared in $|\uparrow\rangle$ and stays at momentum point $\mathbf{k} = (0, 0)$. Four BECs distributed at momenta $\mathbf{k} = (\pm k_0, \pm k_0)$ are in $|\downarrow\rangle$ due to the Raman potential. Here, k_0 is the recoil momentum. Other BECs are in $|\uparrow\rangle$ and diffracted by the lattice potential without spin flipping. (c) Tomography principle. The atoms firstly are in a superposition state $|\Psi\rangle$ of $|\mathbf{q}, \uparrow\rangle$ and $|\mathbf{q} + \mathbf{k}_0, \downarrow\rangle$ (yellow arrows). Applying a $\pi/2$ Raman pulse, $|\Psi\rangle$ evolves to a new superposition state $|\tilde{\Psi}\rangle$ (magenta arrows) along the sky blue trajectories. $\Delta\varphi$ is exactly the angle between the gray axis and the x axis. The gray axis rotates in the equatorial plane by tuning $\Delta\varphi$. After ToF imaging, $\langle\sigma_x\rangle$ ($\langle\sigma_y\rangle$) is obtained when $\Delta\varphi = \pi/2$ (0). $\langle\sigma_z\rangle$ is measured directly by ToF imaging, without applying the Raman pulse. (d) The experimental sequence. The atoms are loaded in Raman lattices in the preparation stage. The Raman pulse is realized by suddenly turning off reflected beams of $E_{x,y}$, setting the Raman pulse strength to Ω_{R0} , the relative phase $\Delta\varphi$ to a certain value, and $\delta_R = 0$. After ToF imaging, the spin polarization P_m at the quasimomentum Γ and M as a function of the relative phase $\Delta\varphi$. The circles with error bars (blue curves) are from experimental data (numerical calculations). The experimental data are fitted by a sinusoidal function (magenta curves). The insets mark the Γ and M points in the FBZ. Parameters: lattice depth $V_0 = 4.0E_r$, Raman potential strength $\Omega_0 = 1.0E_r$, $\delta = -0.2E_r$, $t_R = 10 \mu\text{s}$, and $\delta_R = 0$. The recoil energy $E_r \approx 2\pi \times 3.7$ kHz.

around an axis on the Bloch sphere with frequency Ω_{R0} . For instance, $|\Psi\rangle$ precesses around the y (x) axis in the equatorial

plane when $\Delta\varphi = \pi/2$ (0) [Fig. 1(c)]. After ToF imaging, the expectation values of Pauli matrices $\langle\sigma_{x,y,z}\rangle$ are obtained by the spin polarization [31]

$$\begin{aligned} \langle\tilde{\Psi}|\sigma_z|\tilde{\Psi}\rangle &= \langle\sigma_z\rangle \cos(2\Omega_{R0}t_R) \\ &+ (\langle\sigma_y\rangle \cos \Delta\varphi + \langle\sigma_x\rangle \sin \Delta\varphi) \sin(2\Omega_{R0}t_R). \end{aligned} \quad (3)$$

When $t_R = 0$, $\langle\sigma_z\rangle$ is obtained; when $t_R = \pi/(4\Omega_{R0})$, $\langle\sigma_x\rangle$ ($\langle\sigma_y\rangle$) is extracted by a $\pi/2$ Raman pulse with $\Delta\varphi = \pi/2$ (0) [Fig. 1(c)].

Now, we clarify this technique in the experiment [31]. The experimental protocol is depicted in Fig. 1(d). First, a BEC of ^{87}Rb atoms is adiabatically loaded into the Raman lattices. The BECs condense at Γ or M in the first Brillouin zone (FBZ), which depends on whether the BEC is prepared in $|\uparrow\rangle$ or $|\downarrow\rangle$. Meanwhile, atoms stay at the lowest band of Raman lattices. Second, the $\pi/2$ Raman pulse is switched on within 200 ns to couple the atoms (see Supplemental Material [31] for the sensitivities of the pulse), which is accomplished by simultaneously executing the following manipulations on the beams $\mathbf{E}_{x,y}$: (i) turning off the retroreflective beams of $\mathbf{E}_{x,y}$; (ii) turning on the strength of the Raman pulse to $\Omega_{R0} \approx 3.4E_r$ by tuning the intensity of $\mathbf{E}_{x,y}$; (iii) setting the relative phase $\Delta\varphi$ to a certain value by changing the initial phase of $\mathbf{E}_{x,y}$; and (iv) setting $\delta_R = 0$ by adjusting the frequency of $\mathbf{E}_{x,y}$. Finally, after holding the Raman pulse for a certain duration $t_R = 10 \mu\text{s}$, we measure the atomic numbers $n_{\uparrow,\downarrow}$ using spin-resolved ToF imaging to obtain spin polarization, i.e., $P_m = (n_{\uparrow} - n_{\downarrow})/(n_{\uparrow} + n_{\downarrow})$.

The measured spin polarizations P_m versus the relative phase $\Delta\varphi$ are shown in Fig. 1(e). The spin polarizations at Γ and M are fitted by sinusoidal functions, which demonstrates the superposition between $\langle\sigma_x\rangle$ and $\langle\sigma_y\rangle$ [cf. Eq. (3)]. Thus $\langle\sigma_x\rangle$ ($\langle\sigma_y\rangle$) are extracted from the fittings when $\Delta\varphi = \pi/2$ (0) at Γ and M , marked by diamonds. For comparison, the numerical simulations of the spin polarizations with the same parameters are also plotted in Fig. 1(e); these values are consistent with the experimental data. All these results indicate that Bloch-state tomography in the Raman lattices is achieved.

Next, we explore the geometry and topology with this Bloch-state tomography to directly map out the quantum geometric tensor from the eigenfunction based on measuring the $\langle\sigma_{x,y,z}(\mathbf{q})\rangle$ of Bose gases in the FBZ. In this case, an almost identical procedure to that used in the aforementioned BEC measurement is employed, except that the atomic cloud is cooled to a temperature around 100 nK [31]. Then, the entire lowest band is occupied by a sufficient number of atoms; simultaneously, around one-third of the atoms populate to the higher bands, which reduces the contrast of the spin polarizations, being the same as in Refs. [26,27,33]. Whereupon, we subtract the high-band contribution based on the Bose distribution determined by the numerical calculations to obtain $\langle\sigma_{x,y,z}(\mathbf{q})\rangle$ for the lowest band (see Supplemental Material [31] for details).

The typical normalized $\langle\sigma_{x,y,z}(\mathbf{q})\rangle$ are drawn in Fig. 2(a), which can be divided into two regions with $\langle\sigma_{x,y,z}\rangle < 0$ and $\langle\sigma_{x,y,z}\rangle > 0$. For $\langle\sigma_x(\mathbf{q})\rangle$ ($\langle\sigma_y(\mathbf{q})\rangle$), the two regions are distributed at the top and bottom (left and right) of the FBZ, respectively. For $\langle\sigma_z(\mathbf{q})\rangle$, one of the regions is centered on

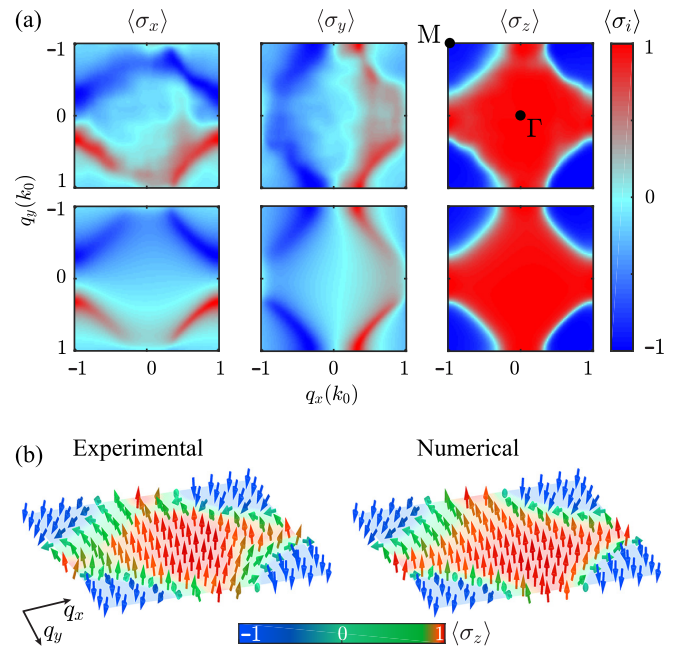


FIG. 2. The expectation values of three Pauli matrices $\langle\sigma_{i=x,y,z}\rangle$ in the FBZ. (a) Normalized $\langle\sigma_{x,y,z}(\mathbf{q})\rangle$ with $\delta = -0.2E_r$ for the lowest band. Results from experimental data (upper row) are compared with numerical calculations (lower row). (b) The distribution of $\langle\sigma(\mathbf{q})\rangle$ in the FBZ takes a skyrmion configuration. Parameters: $(V_0, \Omega_0, \delta) = (4.0, 1.0, -0.2)E_r$.

Γ , and the other region is centered on M for $\delta = -0.2E_r$. The two regions are demarcated by a ring structure with $\langle\sigma_z\rangle = 0$ around M , which is a feature of the band topology in our system [26,27]. The numerical calculations coincide with the experimental measurements. In addition, the distribution of the vectors $\langle\sigma(\mathbf{q})\rangle = (\langle\sigma_x(\mathbf{q})\rangle, \langle\sigma_y(\mathbf{q})\rangle, \langle\sigma_z(\mathbf{q})\rangle)$ shapes a skyrmion [34] structure in the momentum space, and the twisted skyrmion texture is demonstrated in Fig. 2(b). Such a skyrmion configuration offers direct evidence of complete tomography.

Moreover, the eigenfunction can also be extracted from the normalized $\langle\sigma_{x,y,z}\rangle$. To this end, we consider the Bloch Hamiltonian under the two-band tight-binding approximation of Eq. (1) [26,35]. After diagonalizing the Bloch Hamiltonian, the eigenfunction in the lowest band can be written as $|u(\mathbf{q})\rangle = (\sin(\theta_q/2)e^{-i\phi_q}, -\cos(\theta_q/2))^T$, where the parameters $\theta_q = \arccos(-\langle\sigma_z\rangle) \in [0, \pi]$ and $\phi_q = \arg(\langle\sigma_x\rangle + i\langle\sigma_y\rangle) \in [0, 2\pi)$. The amplitudes of the eigenfunction $u_{\uparrow,\downarrow}$ and their relative phase for $\delta = -0.2E_r$ in the FBZ are exhibited in Fig. 3. The amplitude of the eigenfunction $u_{\uparrow}$ has a value of ~ 1 at the Γ point and is close to zero at the M point, whereas the amplitude of $u_{\downarrow}$ is the opposite. The relative phase is significantly different from the amplitude and can be roughly divided into four regions: the blue-gray region ($0 \leq \varphi < 0.5\pi$) and the gray region ($0.5\pi \leq \varphi < \pi$) on the left of the phase map, and the light-green region ($\pi \leq \varphi < 1.5\pi$) and the green region ($1.5\pi \leq \varphi < 2\pi$) on the right of the phase map. At the junctions of these four regions emerge phase vortices, which implies that the vectors $\langle\sigma\rangle$ point to the north or south poles of the Bloch sphere [36].

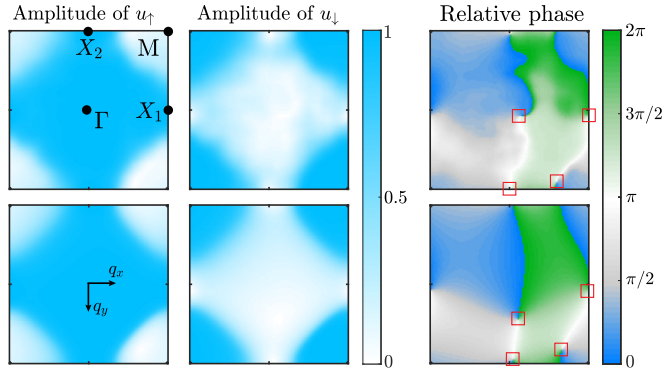


FIG. 3. Experimental reconstruction of the eigenfunction in the FBZ: the amplitude of the eigenfunction $u_\uparrow$ (first column) and $u_\downarrow$ (second column), together with the relative phase between $u_\uparrow$ and $u_\downarrow$ (third column). Red squares mark the positions of the phase vortices. Results from experimental data (upper row) are compared with numerical calculations (lower row). Parameters: $(V_0, \Omega_0, \delta) = (4.0, 1.0, -0.2)E_r$.

Our experimental measurements basically agree with numerical calculations (second row of Fig. 3). Note that the phase vortices deviate from the high-symmetry points of the FBZ $\{\Gamma, M, X_{1,2}\}$, which mainly stems from the coherence between the bands [31].

Thanks to the complete observation of the eigenfunction $|u(\mathbf{q})\rangle$, we can identify the geometric structures of the quantum states by the gauge-invariant quantum geometric tensor $\chi_{\alpha\beta}$ [37]

$$\chi_{\alpha\beta} = \langle \partial_{q_\alpha} u | (1 - |u\rangle\langle u|) | \partial_{q_\beta} u \rangle = g_{\alpha\beta} - i\mathcal{F}_{\alpha\beta}/2. \quad (4)$$

Here, the quantum metric tensor $g_{\alpha\beta} = \text{Re}(\chi_{\alpha\beta})$, as the real part of QGT, measures the distance between quantum states in quasimomentum space [7]. The Berry curvature $\mathcal{F}_{\alpha\beta} = -2\text{Im}(\chi_{\alpha\beta})$, being the imaginary part of QGT, acts as an effective “electromagnetic” tensor in quasimomentum space [5]. Hence the momentum-resolved quantum metric tensor $g_{\alpha\beta}$ and Berry curvature $\mathcal{F}_{xy}$ are extracted from the measurement of eigenfunction in Fig. 3 using Eq. (4), displayed in Fig. 4(a). The Berry curvature $\mathcal{F}_{xy}$ is mainly localized near the ring structure, the sign and direction of which are negative and perpendicular to the plane q_x - q_y , respectively. Thereby, Chern number $\mathcal{C} = \int_{\text{BZ}} \mathcal{F}_{xy} d\mathbf{q}/2\pi = -1.00 \pm 0.02$ at $\delta = -0.2E_r$, signifying that the band is topologically nontrivial. Tuning the detuning from $\delta = -1.0E_r$ to $\delta = 1.0E_r$, in Fig. 4(b), the topological phase diagram is obtained from the Chern number; this phase diagram is consistent with our previous measurements via quench dynamics [35,38,39].

In addition, the quantum metric tensor with nonvanishing value in Fig. 4(a) also appears on the ring structure around M , indicating that the distance between quantum states on the ring is larger than on other areas. Moreover, the integral of the quantum metric tensor over the full Brillouin zone gives a quantum volume [40] $\text{vol}_g = \int_{\text{BZ}} \sqrt{g_{xx}g_{yy} - g_{xy}^2} d\mathbf{q}/\pi$, which yields an inequality $\text{vol}_g \geq |\mathcal{C}|$ for Chern insulators [28,29]. Figure 4(b) shows the quantum volume and the Chern number as functions of the detuning, validating such an inequality experimentally. The inequality is related to superfluidity or

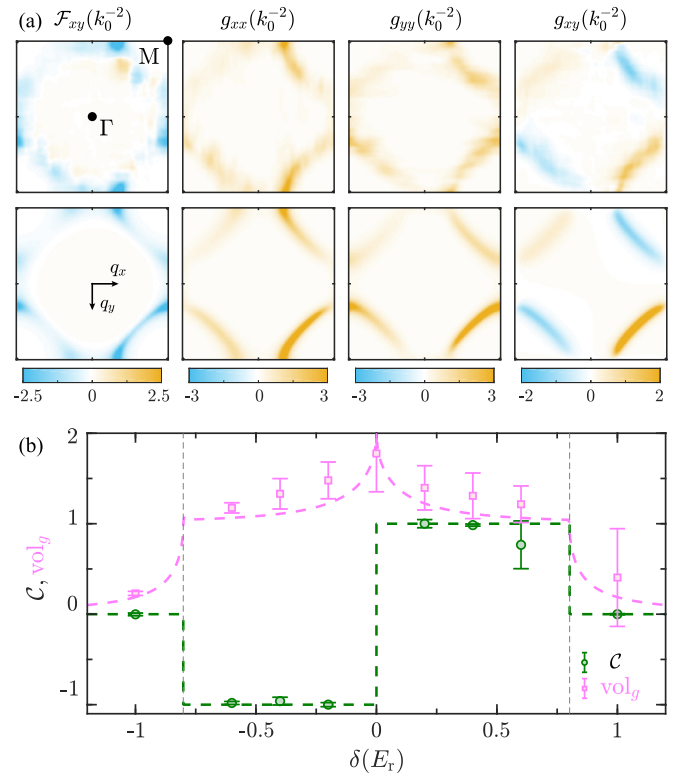


FIG. 4. The quantum geometric tensor in the FBZ. (a) The Berry curvature $\mathcal{F}_{xy}(\mathbf{q})$, together with the three components of the quantum metric tensor, $g_{xx}(\mathbf{q})$, $g_{yy}(\mathbf{q})$, and $g_{xy}(\mathbf{q})$. Results from experimental data (upper row) are compared with numerical calculations (lower row). Parameters: $(V_0, \Omega_0, \delta) = (4.0, 1.0, -0.2)E_r$. (b) The Chern number $\mathcal{C}$ and the quantum volume vol_g vs the detuning δ . The circles and squares with error bars are from experimental results. The dashed pink and green curves are from numerical calculations. Parameters: $(V_0, \Omega_0) = (4.0, 1.0)E_r$.

superconductivity [41]. Thereafter, we may use the inequality to roughly evaluate the topological properties in 2D Raman lattices: If $\text{vol}_g < 1$, $\mathcal{C} = 0$; if $\text{vol}_g \geq 1$ and $\delta \neq 0$, $|\mathcal{C}| = 1$. Note that the quantum volume takes noninteger values that can be straightforwardly interpreted from the positive semidefinite nature of the quantum metric tensor as elaborated in Ref. [42].

Our measurements of the quantum geometric tensor via Bloch-state tomography in 2D optical Raman lattices exhibit a paradigm to investigate the geometry and topology for quantum states. Since such a tomography technique only requires accurate control over the momentum transfer and the phase of the Raman pulse, it can be readily generalized to interacting systems [43], multiband systems [44], and 3D topological systems [33]. It also opens the possibility of spatiotemporal coherent manipulation with a view towards detecting new dynamical topological states of matter. For instance, one could apply the tomography to quench dynamics with the goal of probing the complete structure of the Hopf fibration [21, 45–47]. The quantum metric tensor also provides important information that the Berry curvature is incapable of telling us when the Berry curvature vanishes or diverges, including information about geometric or topological properties [48],

non-Hermitian systems [49,50], geometric orbital susceptibility [51], and Bose and Fermi superfluids [41,52]. Such a situation can be realized by adjusting the parameters of the Raman lattices so that there exist gapless bands, and one can detect certain phenomena in topological semimetals governed by the quantum metric tensor, such as the dynamics of the wave packet [16,49].

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